

TITLE

Air pollution and the risk of bronchiolitis-related hospital admission in a birth cohort of over 400,000 children born in London

SUPPLEMENTARY MATERIAL

Table S1. Adjusted hazard ratios (HR) and 95% confidence interval (CI) for the risk of bronchiolitis-related admission in each landmark period (28 days) of follow-up associated with PM_{2.5}, and NO₂ exposure in the PICNIC London Cohort, 2010-2013.

Table S2 Sensitivity analysis: Adjusted hazard ratios (HR) and 95% confidence interval (CI) for the risk of bronchiolitis-related admission in each landmark period (28 days) of follow-up associated with PM_{2.5}, and NO₂ exposure, using all admissions (emergency and planned) with bronchiolitis recorded as any diagnosis in any episode of the admission) in the PICNIC London Cohort, 2010-2013.

Table S3. Sensitivity Analysis: Adjusted hazard ratios (HR) and 95% confidence interval (CI) for the risk of bronchiolitis-related admission in each landmark period (28 days) of follow-up associated with PM_{2.5}, and NO₂ exposure, restricted to children whose maternal postcode history was available in the PICNIC London Cohort, 2010-2013.

Table S4. Distribution of exposures and number of children who had a bronchiolitis admission or were censored in each landmark period (28 days) of follow-up in the PICNIC London Cohort, 2010-2013.

Figure S1. Directed Acyclic Graph (DAG) for the association between air pollution and bronchiolitis-related admission.

Table S1. Adjusted^a hazard ratios (HR) and 95% confidence interval (CI) for the risk of bronchiolitis-related admission in each landmark period (28 days; LM) of follow-up associated with PM_{2.5}, and NO₂ exposure in the PICNIC London Cohort, 2010-2013.

	LM	HR	95% CI
PM _{2.5} (per 5 µg/m ³)	1	1.07	(0.70, 1.62)
	2	0.97	(0.91, 1.04)
	3	0.93	(0.85, 1.00)
	4	0.95	(0.86, 1.04)
	5	0.99	(0.90, 1.10)
	6	0.89	(0.80, 0.98)
	7	0.86	(0.77, 0.96)
	8	1.00	(0.89, 1.13)
	9	0.98	(0.86, 1.12)
	10	0.97	(0.85, 1.11)
	11	0.97	(0.83, 1.13)
	12	0.90	(0.76, 1.06)
	13	1.12	(0.94, 1.34)
	LM	HR	95% CI
NO ₂ (per 10 µg/m ³)	1	1.15	(0.98, 1.35)
	2	1.00	(0.91, 1.09)
	3	0.98	(0.89, 1.08)
	4	1.08	(0.95, 1.23)
	5	1.01	(0.87, 1.17)
	6	1.00	(0.85, 1.17)
	7	0.99	(0.84, 1.16)

8	1.02	(0.85, 1.22)
9	1.04	(0.85, 1.26)
10	0.90	(0.74, 1.10)
11	0.95	(0.75, 1.20)
12	1.03	(0.81, 1.31)
13	1.31	(1.00, 1.71)

a Models adjusted for year and month of birth, child's ethnicity, maternal age and marital status, index of multiple deprivation (IMD) quintiles, median housing energy efficiency, and average pollution exposures from birth up until the previous landmark period.

Table S2. Sensitivity analysis: Adjusted^a hazard ratios (HR) and 95% confidence interval (CI) for the risk of bronchiolitis-related admission in each landmark period (28 days; LM) of follow-up associated with PM_{2.5}, and NO₂ exposure, using all admissions (emergency and planned) with bronchiolitis recorded as any diagnosis in any episode of the admission) in the PICNIC London Cohort, 2010-2013.

	LM	HR	95% CI
PM _{2.5} (per 5 µg/m ³)	1	0.80	(0.60, 1.08)
	2	0.97	(0.91, 1.03)
	3	0.93	(0.86, 1.00)
	4	0.93	(0.85, 1.02)
	5	1.01	(0.92, 1.12)
	6	0.89	(0.81, 0.98)
	7	0.86	(0.77, 0.95)
	8	0.98	(0.87, 1.10)
	9	0.97	(0.86, 1.10)
	10	0.96	(0.84, 1.10)
	11	1.00	(0.86, 1.16)
	12	0.92	(0.79, 1.07)
	13	1.10	(0.93, 1.30)
	LM	HR	95% CI
NO ₂ (per 10 µg/m ³)	1	0.91	(0.80, 1.21)
	2	1.00	(0.92, 1.09)
	3	0.99	(0.90, 1.09)
	4	1.07	(0.95, 1.22)
	5	1.04	(0.90, 1.21)
	6	1.02	(0.87, 1.18)
	7	0.97	(0.83, 1.14)
	8	0.99	(0.83, 1.18)
	9	1.04	(0.86, 1.26)
	10	0.91	(0.75, 1.11)
	11	1.00	(0.80, 1.26)
	12	1.02	(0.81, 1.28)
	13	1.28	(1.00, 1.65)

a Models adjusted for year and month of birth, child's ethnicity, maternal age and marital status, index of multiple deprivation (IMD) quintiles, median housing energy efficiency, and average pollution exposures from birth up until the previous landmark period.

Table S3. Sensitivity Analysis: Adjusted^a hazard ratios (HR) and 95% confidence interval (CI) for the risk of bronchiolitis-related admission in each landmark period (28 days; LM) of follow-up associated with PM_{2.5}, and NO₂ exposure, restricted to children whose maternal postcode history was available in the PICNIC London Cohort, 2010-2013.

	LM	HR	95% CI
PM _{2.5} (per 5 µg/m ³)	1	1.05	(0.92, 1.11)
	2	1.00	(0.93, 1.07)
	3	0.95	(0.87, 1.03)
	4	0.92	(0.83, 1.02)
	5	0.98	(0.88, 1.1)
	6	0.89	(0.8, 0.99)
	7	0.83	(0.73, 0.93)
	8	1.02	(0.89, 1.16)
	9	0.98	(0.85, 1.12)

	10	0.95	(0.82, 1.1)
	11	0.97	(0.83, 1.15)
	12	0.93	(0.78, 1.11)
	13	1.06	0.87, 1.29)
	LM	HR	95% CI
	1	1.13	(0.94, 1.35)
	2	1.02	(0.93, 1.12)
	3	0.99	(0.89, 1.1)
	4	1.01	(0.88, 1.16)
	5	1.02	(0.87, 1.2)
	6	0.95	(0.8, 1.12)
NO₂ (per 10 µg/m ³)	7	0.98	(0.82, 1.17)
	8	1.05	(0.86, 1.28)
	9	1.06	(0.86, 1.31)
	10	0.87	(0.7, 1.08)
	11	0.92	(0.72, 1.19)
	12	0.99	(0.77, 1.28)
	13	1.32	(0.98, 1.77)

a Models adjusted for year and month of birth, child's ethnicity, maternal age and marital status, index of multiple deprivation (IMD) quintiles, median housing energy efficiency, and average pollution exposures from birth up until the previous landmark period.

Table S4. Distribution of exposures and number of children who had a bronchiolitis admission or were censored in each landmark period (28-day period) in the PICNIC London Cohort, 2010-2013.

Landmark	No. of children	No. of events	No. censored	% Missing	Pollutant	Mean	Std Dev	Min	25th %ile	50th %ile	75th %ile	Max
1	415,311	419	3,807	1.30%	PM _{2.5}	14.5	2.2	4.6	13.2	14.8	15.9	39.6
					NO ₂	31.6	7.4	4.9	26.7	31.1	35.7	113.8
2	411,151	1,757	1,702	0.10%	PM _{2.5}	15.1	4.9	3.8	11.4	14.3	18.2	40.3
					NO ₂	31.5	9.5	5	24.7	30.8	37.4	111.6
3	407,726	1,382	1,353	0.20%	PM _{2.5}	15.1	4.9	3.8	11.4	14.3	18.1	37.1
					NO ₂	31.6	9.6	5	24.7	30.8	37.5	121.4
4	405,018	929	1,088	0.20%	PM _{2.5}	15	5	3.7	11.3	14.3	18.2	39.7
					NO ₂	31.4	9.6	5	24.5	30.6	37.4	135.2
5	403,026	762	1,035	0.30%	PM _{2.5}	14.8	5.2	3.7	11.1	14.2	18.2	43
					NO ₂	30.8	9.4	5	24.1	30.2	36.7	141.3
6	401,247	706	991	0.30%	PM _{2.5}	14.7	5.3	3.6	10.9	14.2	18.3	37.1
					NO ₂	30.5	9.3	5	23.8	29.8	36.3	127.1
7	399,538	572	978	0.30%	PM _{2.5}	14.8	5.4	3.7	10.9	14.3	18.5	35.7
					NO ₂	30.3	9.2	5	23.7	29.6	36.2	121
8	397,994	490	1,001	0.40%	PM _{2.5}	14.7	5.1	3.7	10.9	14.4	18.3	38.3
					NO ₂	30.2	9.2	5	23.6	29.4	35.9	131.2
9	396,509	428	1,119	0.40%	PM _{2.5}	14.4	4.9	3.7	10.9	14.2	17.9	33.1
					NO ₂	29.9	9.1	5	23.4	29.1	35.5	116
10	394,964	365	1,087	0.40%	PM _{2.5}	14.3	4.8	3.7	10.9	14.1	17.6	33.4
					NO ₂	29.7	9.1	5	23.3	28.9	35.3	112.8
11	393,515	310	1,156	0.40%	PM _{2.5}	14.3	4.8	3.7	11	14.1	17.6	32.4
					NO ₂	29.7	9	5	23.3	28.9	35.3	104.7
12	392,053	302	1,208	0.50%	PM _{2.5}	14.2	4.8	3.1	11	14.1	17.6	33.8
					NO ₂	29.7	9	5	23.2	28.8	35.3	127.1
13	390,546	228	1,220	0.50%	PM _{2.5}	14.3	4.8	3.7	11	14.2	17.7	33.5
					NO ₂	29.7	9	5	23.2	28.9	35.3	121

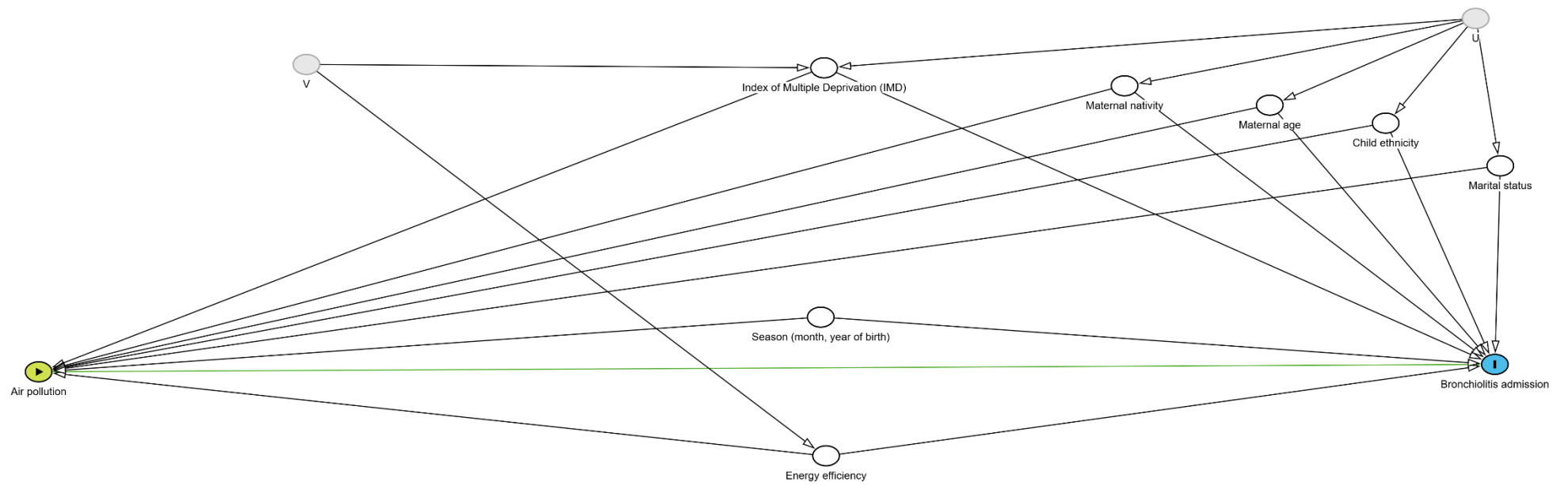


Figure S1. Directed Acyclic Graph (DAG) for the association between air pollution and bronchiolitis admission.